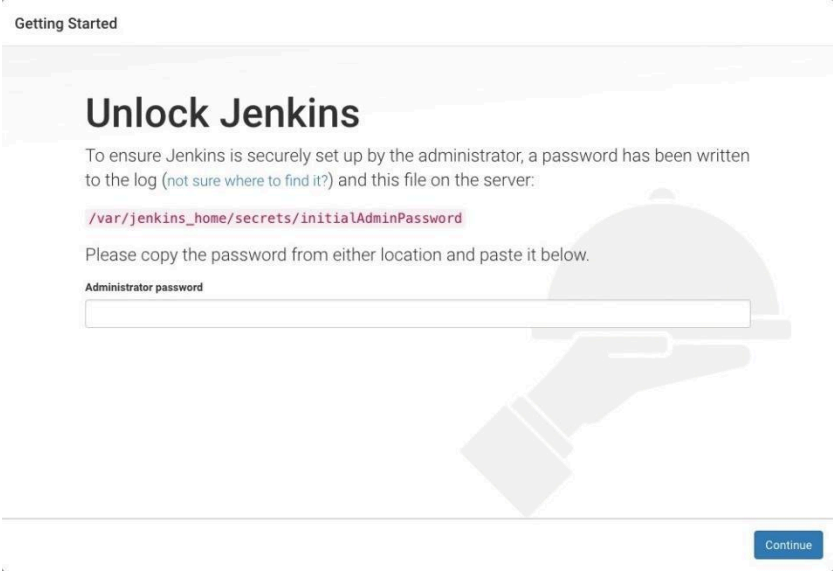


11) Test execution through Jenkins

Install Jenkins in local machine:

- ☐ Search "jenkins download" or go to <https://jenkins.io/download/>
- ☐ Click on .war file to download due to install in local machine.
- ☐ Get the directory/ path of .war file (ex: C:\Users\Jenkins_File)
- ☐ Go to terminal/command prompt window. Type:cd path (C:\Users\Jenkins_File) > java -jar jenkins.war -httpPort=8080
- ☐ Confirm to see message in command prompt as "Jenkins is fully up and running"
- ☐ **Unlocking Jenkins:** Browse to <http://localhost:8080> (or whichever port you configured for Jenkins when installing it) and wait until the **Unlock Jenkins** page appears.
- ☐ Password:



Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`/var/jenkins_home/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

Continue

- ☐ From the Jenkins console log output, copy the automatically-generated alphanumeric password (between the 2 sets of asterisks).

```
INFO: Pre-Instantiating singleton in org.springframework.beans.factory.support.DefaultListableBeanFactory@24cf7404: defining beans [filter,legacy]; root of factory hierarchy
Sep 30, 2017 7:18:39 AM jenkins.install.SetupWizard init
INFO:

*****
*****
*****

Jenkins initial setup is required. An admin user has been created and a password generated.
Please use the following password to proceed to installation:

2f064d3663814887964b68294057250f

This may also be found at: /var/jenkins_home/secrets/initialAdminPassword

*****
*****
*****

--> setting agent port for jnlp
--> setting agent port for jnlp... done
Sep 30, 2017 7:18:51 AM hudson.model.UpdateSite updateData
INFO: Obtained the latest update center data file for UpdateSource default
Sep 30, 2017 7:18:52 AM hudson.model.UpdateSite updateData
INFO: Obtained the latest update center data file for UpdateSource default
Sep 30, 2017 7:18:52 AM hudson.WebAppMain$3 run
INFO: Jenkins is fully up and running
Sep 30, 2017 7:18:52 AM hudson.model.DownloadService$Downloadable load
INFO: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
Sep 30, 2017 7:18:58 AM hudson.model.DownloadService$Downloadable load
INFO: Obtained the updated data file for hudson.tools.JDKInstaller
Sep 30, 2017 7:18:59 AM hudson.model.AsyncPeriodicWork$1 run
INFO: Finished Download metadata. 25,543 ms
```

- ☐ On the **Unlock Jenkins** page, paste this password into the **Administrator password** field and click **Continue**.
- ☐ Create Admin user & Password
- ☐ Confirm to login Jenkins server: <http://localhost:8080/>

Configure Jenkins job to connect with Maven Project:

- ☐ **Create Job:** Click on New Item>Type item name (ex: Codenbox WebAutomation)>
- ☐ Select Free style project>Ok>
- ☐ Select 'Source Code Management' is None
- ☐ Click Advance under general>Select 'use custom workspace'>copy the project's url and past the link under Directory
- ☐ scroll down & select 'invoke top level maven targets' under Build
- ☐ Under Goals: test -Pgroups (mvn is not required) >Save
- ☐ Click on Build Now
- ☐ Check the console for output log

Make Jenkins Project Parameterized:

- ☐ Go to project's Configure
- ☐ Check off 'This project is parameterized' option

- ☐ Select 'Choice Parameter' under 'Add Parameter'
- ☐ Give a parameter name (ex: Profile)
- ☐ Include ' master' + 'groups' (whatever profile want to add as per profile id in POM) under Choices
- ☐ Update Goals under Build as test -P"\$Profile" >Save
- ☐ Build with Parameter & select Profile want to execute >Build

Integrate TestNG XML report:

- Login to Jenkins>Manage Jenkins> Manage Plug in>Search 'testNG Result' under available tab>Select TestNG Result>Install >restart>Login back
- ☐ Go back to the Project in Jenkins>configure>Select 'Publish TestNG Results' under 'Post-build Actions'>Save

Integrate Extent Reports to Jenkins project:

- Open jenkins server: java -jar jenkins.war -httpPort=8080
- Make sure to delete the TestNG Report
- Login to Jenkins>Manage Jenkins> Manage Plug in>Search 'html publisher' under available tab>Install >restart>Login back
- Go back to the Project in Jenkins>configure>Select 'Publish HTML reports' under 'Post-build Actions'>
- Click on Add
- Add 'extent report' director path to HTML directory
- Add .html to index page
- Change Report title to 'Extent Report'
- Apply & Save
- Start build and check the extent report
- If you find the report shows only raw data then follow the below steps:
 - Terminate jenkins server (ex: Ctrl+c)
 - Rerun jenkins server with : **java -Dhudson.model.DirectoryBrowserSupport.csp="" -jar jenkins.war -httpPort=8080**
 - Go to manage Jenkins > Enter to Script Console

```
System.setProperty("hudson.model.DirectoryBrowserSupport.CSP",  
"default-src 'self'; style-src 'self' 'unsafe-inline';")
```

OR

```
System.setProperty("hudson.model.DirectoryBrowserSupport.CSP",  
"default-src * 'unsafe-inline' 'unsafe-eval'; script-src * 'unsafe-inline'  
'unsafe-eval'; connect-src * 'unsafe-inline'; img-src * data: blob:  
'unsafe-inline'; frame-src *; style-src * 'unsafe-inline';")
```

- Back to project and build to execute the test
- Check Console output and Extent Report
- Should able to see the details report including the past one

Scheduling Jenkins Jobs for a specific time: To setup an schedule for a job/project due to build and execute the test we need to update our 'Build Triggers' in Jenkins. Go to the Project>Configure>Select 'Build Periodically' under Build Triggers and update/type our schedule (ex: 30 2 * * *)as per we need as bellow process and Save:

Jenkins uses a [cron expression](#), the different fields are :

1. MINUTES Minutes in one hour (0-59)
2. HOURS Hours in one day (0-23)
3. DAYMONTH Day in a month (1-31)
4. MONTH Month in a year (1-12)
5. DAYWEEK Day of the week (0-7) where 0 and 7 are sunday

Since a few versions, Jenkins add a new parameter, H : (extract from the [jenkins code documentation](#))

To allow periodically scheduled tasks to produce even load on the system, the symbol H (for “hash”) 0 (zero) and * should be used wherever possible.

Note :

* means every day/every month/every year
0 means Zero th minute/hour
H means any or between time

Examples 1: The following are the main 5 parameters to set an schedule:

M H D M DAYWeek

0 7 * * * (The job will run 0th minute at 7am every day/month/year)

0 0 * * * (The job will spike 0th minute and 0th hour, means at midnight every day)

30 2 * * * (The job will run 2.30am on every day/month/year)

Example 2 : H H(3-4) * * * : A job which runs every Day of the week, every Month in a year , every Day in a month at a time between 3 – 4 am in the morning at any minute.

Example 3 : H (30 -45) 3 * * * : A job which runs every Day of the week, every Month in a year , every Day in a month at 3 am in the morning between 30 -45 minutes.

Example 4 : */5 * * * * : If you want to schedule your build every 5 minutes, this will do the job

Example 5 : 0 8 * * * : schedule your build every day at 8h 00, this will do the job

More info link:

<https://itisatechiesworld.wordpress.com/jenkins-related-articles/jenkins-configuration/jenkins-scheduling-jenkins-jobs-for-a-specific-time/>

Run test script in headless:

- ☐ Go to Base class and update setup () method by adding a case option

case "chromeheadless":

ChromeOptions options=new ChromeOptions();

```
options.addArguments("headless");
driver = new ChromeDriver(options);
driver.manage().window().setSize(new Dimension(1440,
900));

break;
```

- ☐ Run the project: `mvn test -Pgroups -DBrowser=chromeheadless`

Make browser Parameterized in Jenkins:

- ☐ Go to project's Configure
- ☐ Check off 'This project is parameterized' option
- ☐ Select 'Choice Parameter' under 'Add Parameter'
- ☐ Give a parameter name (ex: Browser)
- ☐ Include chrome, chromeheadless, firefox, edge (all browser list want to add as per Base class options) under Choices
- ☐ Update Goals under Build: test -P"\$Profile" -DBrowser="\$Browser"
- ☐ Build with Parameter & select Profile & Browser want to execute

****🚀 Let's Connect ▼ ****

👉 Youtube▶ : @codenboxAutomationLab

👉 LinkedIn▶ : <https://.linkedin.com/in/sarifuli/>

✅ Looking forward to connecting with you all and making a difference together! 🙏🚀